

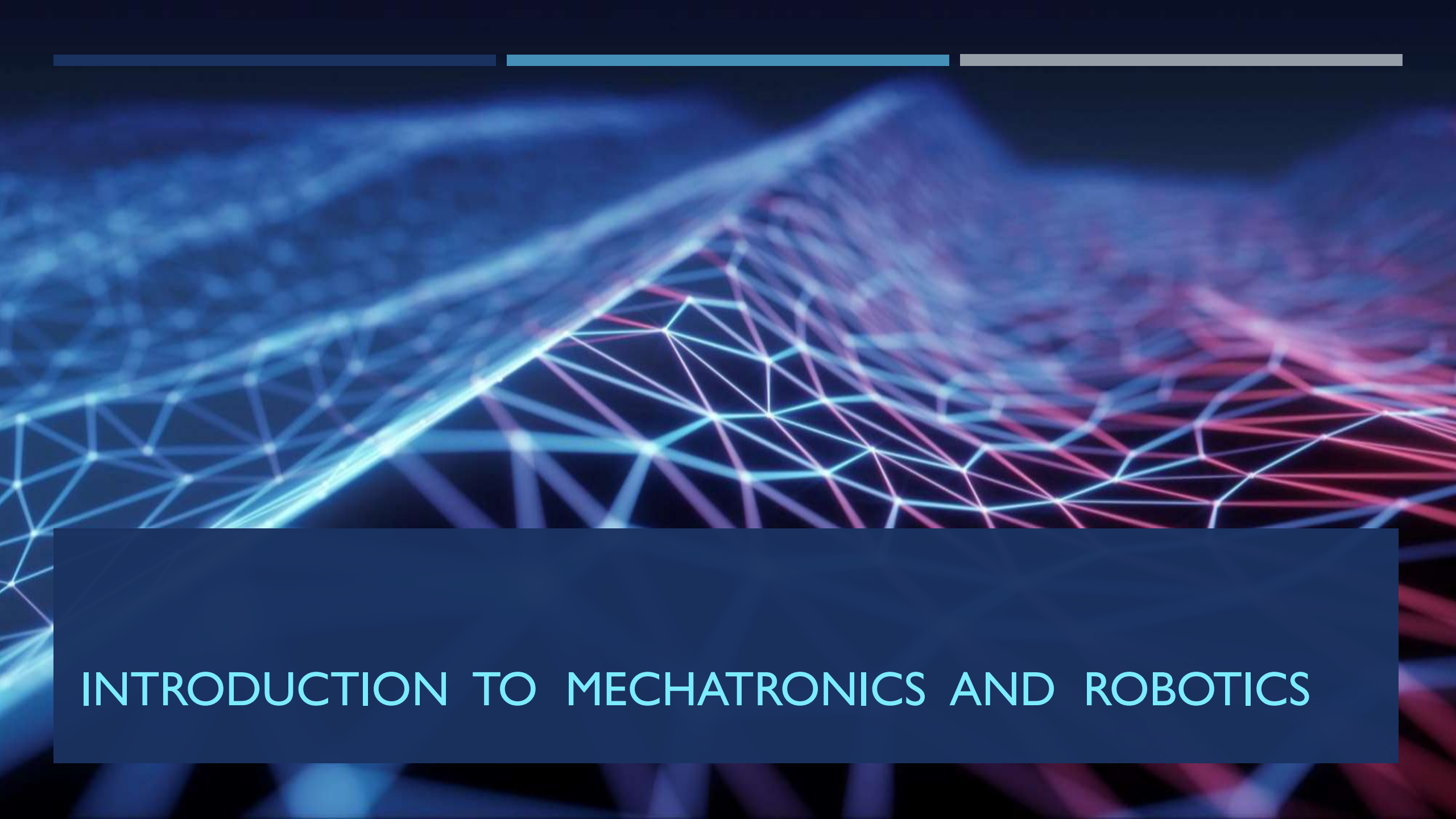


# JNANA VIKAS INSTITUTE OF TECHNOLOGY

## PROJECT REPORT ON INTRODUCTION TO MECHATRONICS AND ROBOTICS

Submitted to :  
Prof . Puneeth.S  
Submitted by :  
Yashmitha .V

From :  
Branch : ISE  
Cycle : physics  
USN : 1JV24IS037



# INTRODUCTION TO MECHATRONICS AND ROBOTICS

# MECHATRONICS :

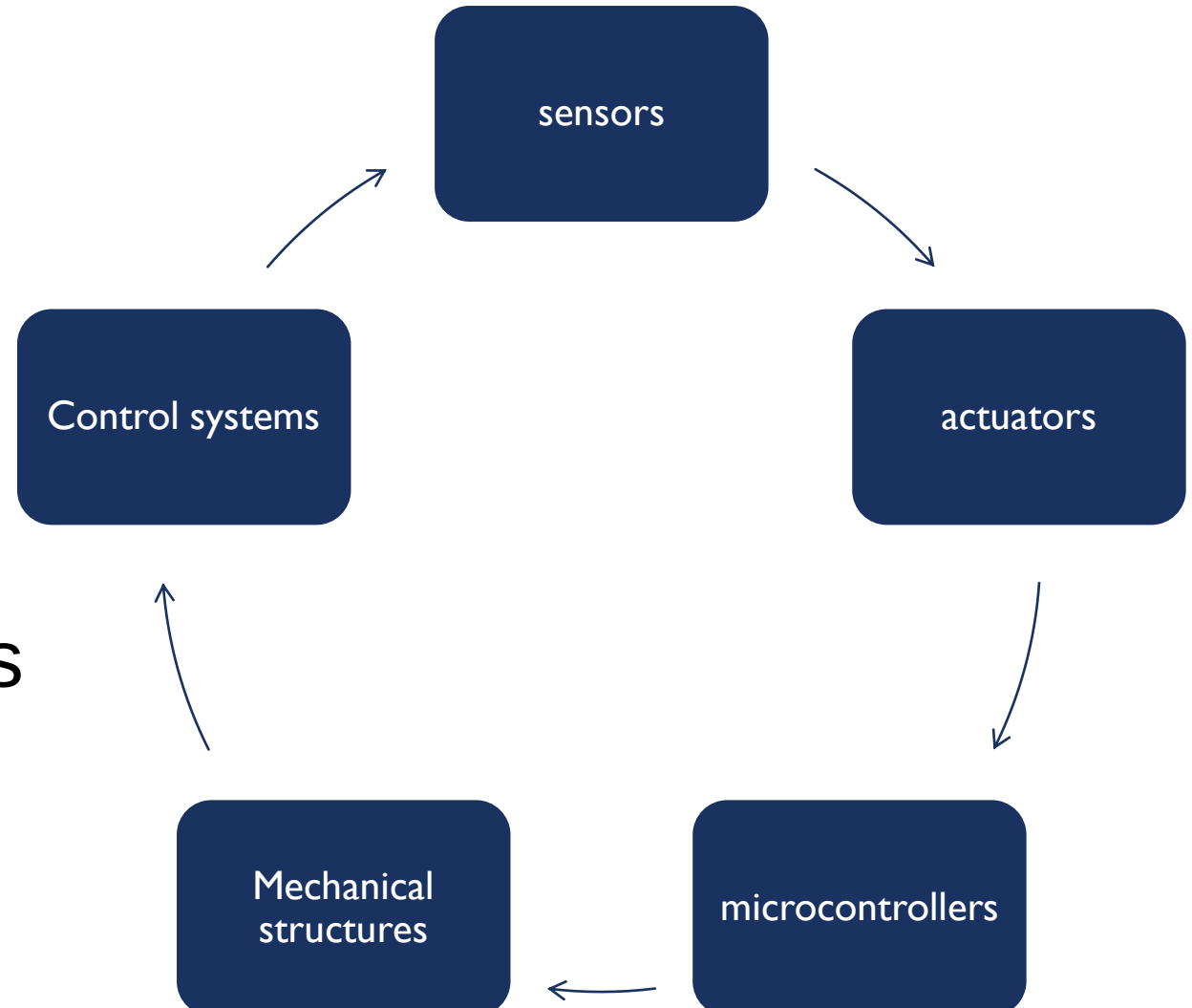
**Mechatronics is a multidisciplinary field that refers to the skill sets needed in the contemporary, advanced automated manufacturing industry.**

- **At the intersection of mechanics, electronics, and computing, mechatronics specialists create simpler, smarter systems.**
- **Mechatronics is an essential foundation for the expected growth in automation and manufacturing.**
- **Mechatronics deals with robotics, control systems, and electro-mechanical systems**

# COMPONENTS OF MECHATRONIC SYSTEMS

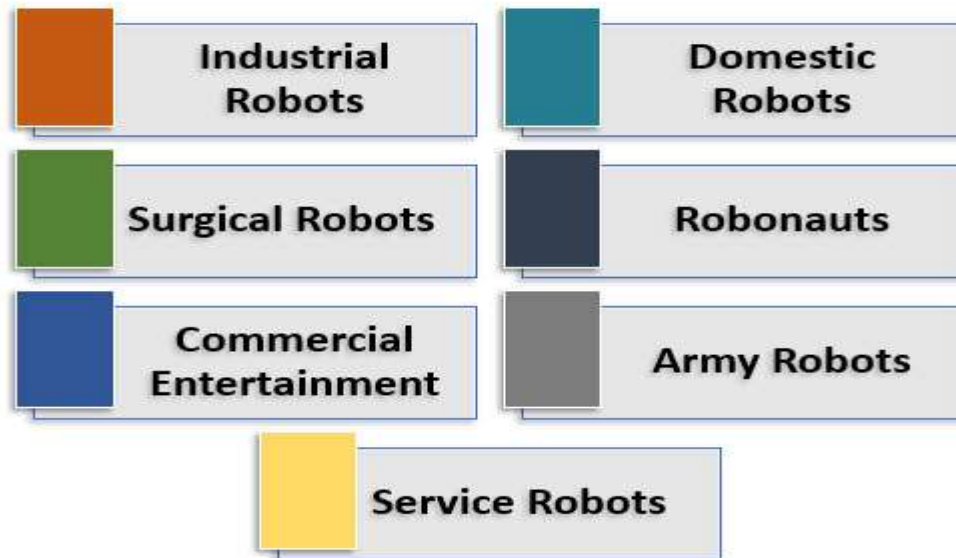
- **Main components:**

- Sensors
- Actuators
- Microcontrollers
- Mechanical structures
- Software/Control systems



# ROBOTICS

Robotics is the branch of technology that deals with the design, construction, operation, and application of robots.



# DIFFERENCE BETWEEN MECHATRONICS AND ROBOTICS

Mechatronics

Broader field

Includes non-robotic systems

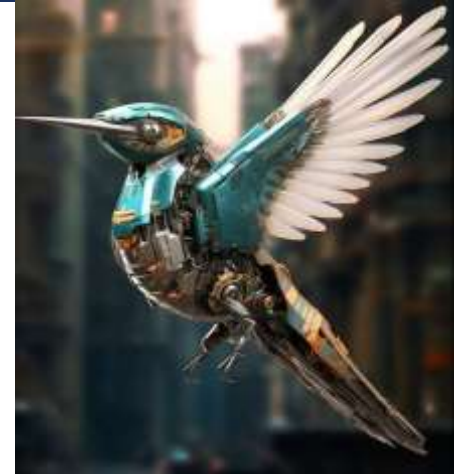
Examples: Washing machine, ABS system

Robotics

Subset of mechatronics

Focuses only on robots

Examples: Humanoid robot, robot vacuum



# APPLICATIONS OF MECHATRONICS AND ROBOTICS

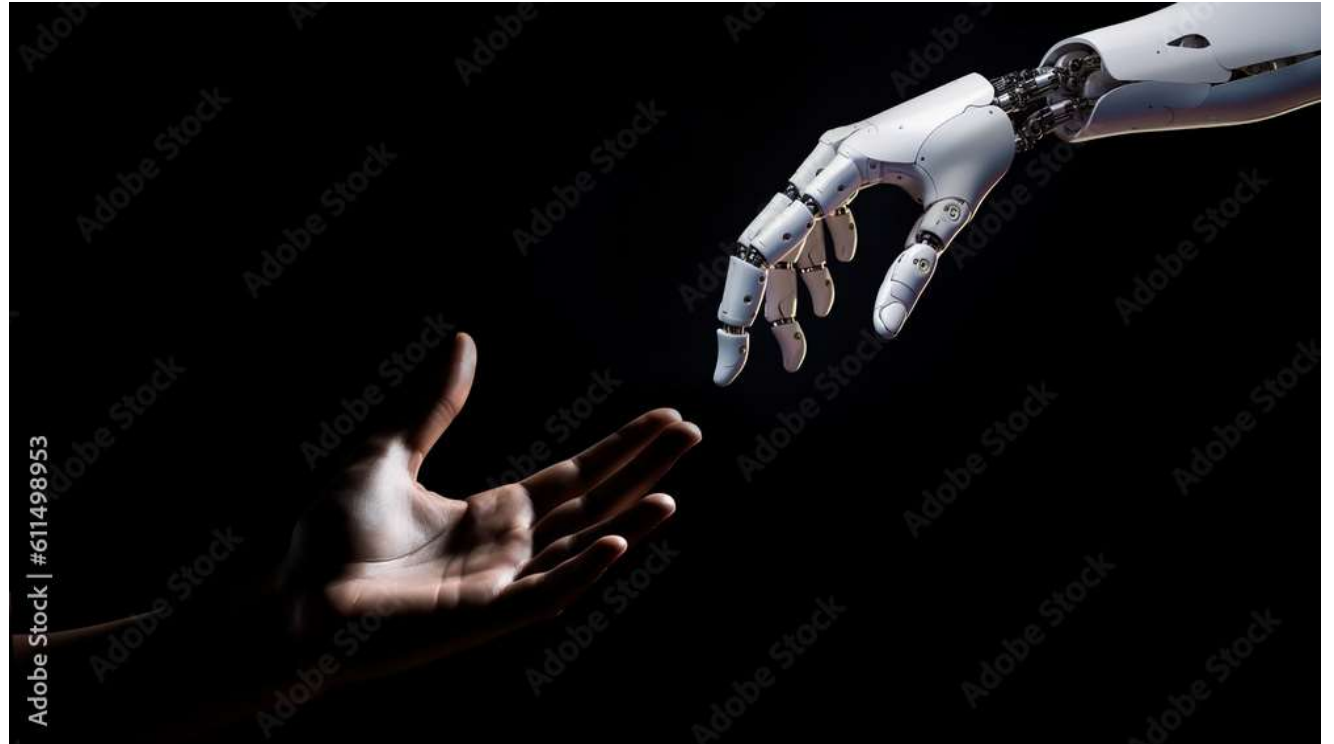
## Examples:

- Industry (automation)
- Healthcare (surgical robots)
- Military (drones)
- Agriculture (robotic harvesters)
- Home (smart appliances)



# FUTURE OF MECHATRONICS AND ROBOTICS

- AI integration
- Human-robot collaboration
- Autonomous vehicles
- Space exploration
- Smart cities





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# THANK YOU

YASHMITHA V

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